NANOPTIC

Lab Foundations

Version 1.1 · 30 October 2026

How Nanoptic Research works: what the lab is, how it runs, how it earns trust, where it is going, and how it does research.

Director: Esethu (Seth) Tshintsholo

Supersedes Lab Foundations v1.0 (28 October 2026). Amendment v1.1 approved 30 October 2026.

What changed in v1.1

Version 1.1 keeps everything in v1.0 except the changes below. Changed or new text in this document is marked **[v1.1]**. The full amendment is in Section 9.

Area	v1.0	v1.1
Name	Unnamed ("the lab")	Nanoptic: Nanoptic AI / Nanoptic Research (na-NOP-tic)
Base model	None	Pico: a small world model, built and trained from scratch
Study IDs	LAB-YYYY-NNN	NAN-YYYY-NNN
Study plan form	Study Card	Research Card
Purpose	Teaching Seth and doing research, mixed	Research lab only; Seth's maths roadmap runs separately
Who writes code	Seth, line by line	Claude Code; Seth explains every block in plain English
Checking	Verification checklist	Checklist plus a second, independent check of every key result

1. What the lab is

[v1.1] Nanoptic (Nanoptic AI / Nanoptic Research, pronounced na-NOP-tic) is a one-person research lab. Its job is to understand how AI works, test ideas honestly, and share what it finds. Success is measured in understanding gained, not money earned.

Research focus: how AI can build an internal model of the physical world (world models), and how to do that with far less energy, taking ideas from how the brain works.

[v1.1] Pico is the lab's base model: a small world model that Nanoptic designs, builds and trains from scratch. Open-source models may be used as tools or baselines, never as Pico's foundation.

The lab DOES:

- **[v1.1]** run real, small, honest research on free cloud computers
- write down everything it tries, including failures
- publish what it learns in plain words, with code
- **[v1.1]** keep its research separate from Seth's personal maths learning, which runs alongside it (Section 2)

The lab does NOT:

- chase money, customers or a product
- compete with school
- overclaim
- force a schedule that causes burnout (missed days are fine, quitting is not)

Scale today: one researcher, a weak laptop, a phone, free cloud GPUs, no budget.

[v1.1] Roles.

Who	Owns
Seth (director-researcher)	Research questions, plain-English specs, predictions, interpreting results, final claim wording
Claude Code	Writing the code from Seth's specs and the workorder
Claude (Research Partner)	Explaining, reviewing, designing checks, catching errors, arguing the strongest case against conclusions

2. Daily operating structure

[v1.1] Daily loop: **Plan > Learn > Research > Log > Rest.** The Learn block is Seth's personal maths roadmap and sits outside the lab. The Research block is one step of lab work. About 60 minutes on school days, up to 2–3 hours on free days. Homework and test study always come first.

1. Plan (5 min): open yesterday's log; its "Next step" line is today's task.
2. **[v1.1]** Learn (25 min school day / 60 min free day): one roadmap row, one course chapter, or one paper section. Take notes. This is personal study, not lab output.
3. **[v1.1]** Research (20 min / 60–90 min): one lab step, such as writing a spec, reviewing what Claude Code built, running a planned experiment, or interpreting a result.
4. Log (10–15 min): fill in the Daily Log. Always write the "Next step" line.
5. Rest: stop. Unfinished work is fine.

Rules

- One task per day (split big tasks).

- **[v1.1]** Never run code whose job you can't explain in plain English, block by block. (Replaces v1.0's "line by line" rule.)
- Log every session, even bad ones.
- One full rest day per week.
- Pause the lab during exams with a one-line log entry.

Task types

Type	What happens
Study day	**[v1.1]** One roadmap subtopic to its stop test (personal, outside the lab)
Build day	**[v1.1]** Seth writes a plain-English spec; Claude Code builds; Seth explains each block back
Paper day	Three-pass reading: 1 title, abstract, figures; 2 full text skipping proofs; 3 work through the method
Experiment day	One step of an active study
Write day	Turn the week's logs into the weekly note

Sample term week: Mon Study · Tue Build · Wed Study · Thu Paper or experiment · Fri Build or experiment · Sat Weekly note + weekly review · Sun Rest.

3. Weekly credibility guide

Credibility comes from public work anyone can check, published steadily for a long time.

Every week

1. Publish one weekly note on Saturday (300–800 words: learned, built, failed, next; code linked).
2. Push all code to GitHub; every experiment folder has a README explaining how to re-run it.
3. Do the 15-minute weekly review and pick next week's focus.
4. Engage usefully with one other researcher's work (comment, question, issue).
5. Correct mistakes in public with a dated note; never delete quietly.

Every month

One deeper piece (a paper reproduction, an explainer of a hard concept, or a small study written up end to end) and an update to the lab's index page.

Every quarter

One bigger output (a Kaggle or Zindi competition entry, an evaluation of open models on one question, or a small dataset collected and cleaned), and ask one person to review a piece of work. **[v1.1]** Because AI writes the code, this human review matters more than before.

Builds trust	Destroys trust
Code anyone can re-run; showing failures and limits; claims matched to evidence; steady output for a year; dated corrections; crediting every source and tool, including AI	Results without code; only showing wins; hype words ("revolutionary", "breakthrough"); bursts then silence; quiet edits; hiding AI use

Audience: don't chase followers. Publish on GitHub, a simple blog, and short summaries linking back. Count understanding, not likes.

4. Vision

> **Vision statement (draft):** to help give machines a real understanding of the physical world, through world models that learn more like the brain does and use far less energy, so that robots can perceive, predict and act safely in everyday life.

Path: a direction, not a deadline. Each stage starts only when the previous one is genuinely done.

1. **Foundations** (now to end of Grade 12): roadmap Phases 1–7, school physics, Python. **[v1.1]** Lab outputs: Pico's test bench, the first versions of Pico, weekly notes.
2. **First world-model experiments** (from about Phase 7): small webcam experiments, e.g. predicting the next video frame or learning simple physics from video. Outputs: reproductions of small known papers, honest write-ups.
3. **University and deeper research:** join a university lab, take research projects, read the core literature on world models, robotics and neuromorphic computing. Outputs: first papers or preprints.
4. **Specialisation:** one sharp question inside world models for robotics (e.g. world models on low-power hardware). Outputs: a recognised body of work and collaborators.
5. **Optional, lab to company:** only if a result solves a real problem others need solved. A company may appear here or never; either is acceptable.

Never changes: honest claims, public work, steady pace, school and health first.

5. Motto

Core values: Curiosity first · Learn from nature · Real-world impact. **Tone:** rigorous and precise.

> *"We research because we want to understand, not because we need an outcome. Our first teacher is nature: the brain shows that real intelligence can perceive, predict and act on very little energy, and we study it to learn how machines might do the same. We work with discipline: every question is stated exactly, every method is written down before the result, and every claim is measured against what was actually shown. We measure our work by one test only, whether it brings machines closer to understanding the real world in a way that helps the people who live in it."*

Candidate one-liners (not final): "Understand first." · "Learn from nature. Build for the world." · "Curious by nature, rigorous by method."

6. Research protocol

A one-person version of the Naple Research Protocol (NRP v1.0, 22 Oct 2026). Core: ask a precise question, write the plan before the results, record everything, check it, say no more than the evidence allows, correct mistakes in the open. For lab work, this protocol takes precedence over the full NRP.

When to use it: build and study days need only the Daily Log. Any experiment that will be published as a finding goes through the full lifecycle and gets an ID **[v1.1]** (**NAN-YYYY-NNN**, e.g. NAN-2026-001).

Study lifecycle

1. **Question:** what will be measured, under what conditions, compared to what. ("Is my model good?" fails. "Does a second hidden layer cut MNIST test error by at least 1 percentage point, over 3 runs each?" passes.) Fill in the **[v1.1] Research Card**. Question types: existence, size, comparison, mechanism, feasibility.
2. **Survey:** 1–3 sessions finding what's known; separate established, contested and open. If it's already answered, reproduce it instead or pick a new question.
3. **Plan and freeze:** before running anything, write the prediction, method, baseline, number of runs, and the result that would prove you wrong. Commit the plan to GitHub; the commit date proves it came first.
4. **Run and record:** follow the plan; log every run in the Experiment Record; any change goes in Deviations the same day with the reason.
5. **Analyse:** report the planned analysis first, whatever it shows. Anything else is labelled exploratory.
6. **Verify:** run the checklist; re-run everything from a clean notebook using only the written record. **[v1.1]** Every key result is checked a second, independent way.
7. **Self-review:** write a "strongest case against" section; ask someone else to read it if possible.
8. **Publish:** claim words match the evidence ladder; include code, frozen plan, deviations, limitations and the AI-use statement. **[v1.1]** Before publishing, Seth must be able to explain the result, the method and the weaknesses in plain English (study literacy).
9. **Archive:** keep repo, data links and logs; add the study to the lab index with its final status.

Machine-learning rules

- Never tune on the test set. Split train / validation / test before starting; look at test once, at the end.
- Always compare to a baseline (a simple model or a published number).
- At least 3 seeds; report mean and spread (standard deviation or range).
- Fix and record seeds, library versions and hardware (GPU type, run time).
- Every figure comes from a script; never edit charts by hand.
- Small data, small claims.

Verification checklist (adapted from NRP §10.3)

[ ] Sanity: beats random guessing and a trivial baseline.
[ ] Shapes and units are as expected.
[ ] Limiting cases: zero learning rate changes nothing; a tiny dataset can be memorised.
[ ] Independent route: key number obtained a second way (e.g. numerical gradient check).
[ ] Clean re-run from the written record reproduces the result within the reported spread.
[ ] Benchmark against a published result where one exists; gap explained.
[ ] No leakage: test data never used for training or tuning.
[ ] Every number carries a source label.

Source labels (simplified from NRP §13)

Label	Meaning
MEASURED	We ran it; code and seed recorded
DERIVED	Worked out by maths; derivation shown
CITED	From a paper or official source; full reference
ESTIMATED	Our rough estimate; method stated
ASSUMED	A chosen value; reason stated
UNVERIFIED	From memory or an AI, not yet checked. NEVER in a published piece

Evidence ladder (from NRP §14)

Level	Meaning	Strongest words allowed
E0 Conjecture	Idea, untested	"I conjecture"
E1 Plausible	Survey done, nothing contradicts it	"is plausible"
E2 Indicated	Exploratory or single-run result	"suggests"
E3 Supported	Pre-planned, 3+ seeds, spread reported, re-run passed	"I find"
E4 Established	Reproduced by someone else, or published with data	"is established"

AI use

AI (including Claude and Claude Code) can explain, suggest and write code; everything it produces is UNVERIFIED until checked. **[v1.1]** Claude Code writes Nanoptic's code. Because no human checks that code line by line, every key result needs a second, independent check, and claims stay small. Every published piece ends with an AI-use statement: which AI, for what, and how it was checked.

7. Templates

Daily Log

```
Daily Log — YYYY-MM-DD
Task type: Study / Build / Paper / Experiment / Write
Time spent: __ min
Goal today: (one sentence, from yesterday's Next step)
What I did:
What I learned: (in my own words)
What confused me / failed:
Links: Code | Notes | Sources
Next step: (one concrete task)
```

[v1.1] Research Card (was "Study Card")

```
Research Card — NAN-YYYY-NNN: title
Date opened:
Question: (what is measured, conditions, compared to what)
Question type: existence / size / comparison / mechanism / feasibility
Why it matters:
Already answered? (survey summary + links)
Prediction: (a number with a range)
Proves me wrong if:
Baseline:
Dataset + split:
Method: (steps)
Runs: (number of seeds)
Analysis plan:
Stop / kill criteria:
Plan frozen: (GitHub commit link + date)
```

Experiment Record (one entry per run)

```
Run N — YYYY-MM-DD
Study: NAN-YYYY-NNN
Code: (commit link) | Notebook:
Seed: | Hardware: | Run time:
Settings changed from plan: none / (describe)
Result: (numbers with labels)
Observations:
Deviations: what changed / why / effect / exploratory now? Y/N
```

Weekly Review (private, Saturday)

```
Weekly Review — week of YYYY-MM-DD
Sessions done: _/6
Biggest thing I understood:
What I built:
Where I got stuck:
Roadmap progress: (phase + subtopic)
Energy for the lab (1–5):
Next week's focus: (one theme)
Published this week: (link)
```

Weekly Note (public, Saturday)

```
Week N: (headline of what I learned)
TL;DR: (two sentences)
What I worked on / What I found (with evidence level) / What didn't work /
What I'm still unsure about / Next week
Code: | Sources: | AI use: | Corrections (dated):
```

Study Write-up

Weekly Note shape, plus: frozen plan link, all deviations, verification checklist results, "strongest case against", limitations, and the evidence level of each main claim.

8. Toolkit

Where	Tools
Laptop	VS Code; Python with NumPy, Matplotlib and Jupyter; Git and GitHub; **[v1.1]** Claude Code; Google Colab and Kaggle (free GPUs; check current limits); Obsidian (lab notebook); Zotero (papers and citations); Hugging Face account; a simple blog (GitHub Pages or Substack)
Phone	Obsidian mobile, GitHub mobile, a PDF reader with highlighting, Anki (formula flashcards), camera for handwritten notes. (Phone type and sync setup not yet confirmed.)

Repo structure

```
nanoptic/
  README.md        what the lab is + index of all work
  logs/
  notes/
  studies/
    NAN-YYYY-NNN/
      card.md        Research Card
      WORKORDER.md   instructions for Claude Code
      records.md     Experiment Records
      src/ tests/ results/ figures/
      writeup.md
  weekly/
  templates/
```

[v1.1] Studies may use scripts instead of a single notebook, as long as every result can be re-run from the written record. Large data and model files live on Google Drive or Hugging Face, linked, never in the repo.

9. Amendment log

Amendment v1.1 — approved by Seth, 30 October 2026

1. Nanoptic is a research lab. Seth's personal maths learning (the roadmap) runs separately, outside the lab.
2. Roles: Seth is director-researcher. He owns questions, plain-English specs, predictions, interpreting results and claim wording. Claude Code writes the code. Claude (Research Partner) reviews, designs checks and argues against conclusions.
3. The rule "never run code you can't explain line by line" becomes "never run code whose job you can't explain in plain English, block by block."
4. Study literacy: before publishing a study, Seth must be able to explain its result, its method and its weaknesses in plain English.
5. Every key result is checked a second, independent way before it's published.
6. Reason for the change: the original design mixed teaching and research; separating them keeps both sustainable.

Decisions recorded at the same time (30 October 2026)

- The lab is named Nanoptic (Nanoptic AI / Nanoptic Research), pronounced na-NOP-tic. Logo pack adopted.
- The lab's base model is Pico: a small world model built and trained from scratch.
- Study IDs use the prefix NAN (NAN-YYYY-NNN). First study: NAN-2026-001, Pico's test bench.
- "Study Card" is renamed "Research Card".

Earlier version: Lab Foundations v1.0, 28 October 2026. v1.0 remains the historical record; this document replaces it for all new work.